Chapter 1 Distribution and Derivatives

1.1 Motivation and Overview

1.1.1 Introduction

Take the following notion:

$$C^m(I) = \{u : I \to \mathbb{C} | \frac{d^j}{dx^j} u \text{ exists and is continuous on } I, 0 \le j \le m\}$$

absolutely continuous functions:

...

1.2 Existence of Test Functions

Definition 1.2.0.1

- For an open sest X in $\mathbb{R}^n$, we shall denote by $C^k(X)$ the space off k times continuously differentiable complex valued functions in X if k is a non-negative integer.
- Take the notion of

$$C^\infty(X) := \cap C^k(X)$$

- Take the following notion

$$supp(u) := \overline{\{x | f(x) \neq 0\}}$$

- The test functions: Take the notion of

$$C_0^k(X) := \{u \in C^k(X) | supp(u) \text{ is compact}\}$$

1.2.1 Convolution

Definition 1.2.1.1 If u and v are in $C(\mathbb{R}^n)$ and either one has compact support, then the convolution $u * v$ is the continuous function defined by

$$(u * v)(x) := \int u(x-y)v(y)dy$$

1.3 Cutoff Functions and Partitions of Unity

▶ ***Theorem*** If X is an open set in $\mathbb{R}^n$ and K is a compact subset, then one can find $\phi \in C_0^\infty(X)$ with $0 \le \phi \le 1$ so that

$$\phi(x) = 1, x \in U \supseteq K, U \text{ is open}$$

∘ **Proof**

□

▶ ***Theorem*** Let X be an open set in the N dimensional vector space V with norm $\| - \|$, and K be a compact subset. If

$$d = inf\{\|x - y\| | x \in V - X, y \in K\}$$

∘ **Proof** ... □

◇ **Lemma 1.3.0.1** If K is a convex symmetric body in the n dimensional vector space, then one can find an **ellipsoid** B with center at 0, and

...

∘ **Proof** ... □

▶ ***Theorem*** Let $X_1, \dots, X_k$ be open sets in $\mathbb{R}^n$, and let $\phi \in C_0^\infty(\cup_1^k X_j)$. Then one can find

$$\phi_j \in C_0^\infty(X_j), j = 1, \dots, k$$

If $\phi \geq 0$ one can take all $\phi_j \geq 0$.

∘ **Proof** ... □

▶ ***Theorem*** Let $X_1, \dots, X_k$ be open sets in $\mathbb{R}^n$ and K a compact subset of $\cup_1^k X_j$.
Then, one can find $\phi_j \in C_0^\infty(X_j)$ so that $\phi_j \geq 0$ and

$$\sum \phi_j \leq 1$$

with equality in a neighborhood of K.

∘ **Proof** ... □

▶ ***Theorem*** For every neighborood X of the cube

...

one can find a non-negative function $\phi \in C_0^\infty(X)$ such that

...

∘ **Proof** ... □

Chapter 2 Definition and Basic Properties of Distributions

2.1 Basic Definition

Chapter 3 Differentiation and Multiplication by Functions

Chapter 4 Convolution

Chapter 5　Distibution in Product Spaces

Chapter 6   Composition with Smooth Maps

Chapter 7 The Fourier Transformation

Chapter 8 Spectral Analysis of Singularities

Chapter 9 Hyperfunctions